

Eric Gulotty Jr.

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WORK EXPERIENCE

MITRE | Bedford, MA & McLean, VA

May 2024 - Present

Intermediate Data Scientist

- Engineered an AI-enabled analysis dashboard (Python/Vue.js) pairing LLMs with simulation models and SME-driven prompt engineering to enable non-SME analysts to perform SME-level analysis and accelerate acquisition timelines; selected as pilot lead scaling the platform across classification levels and program offices, with RAG and knowledge-management expansion underway and onboarding discussions active with organizations focused on rapid prototyping of emerging technology.
- Built a firm-level forecasting model to quantify the economic effects of proposed tariff schedules, integrating diverse datasets to support rapid scenario analysis for federal stakeholders.
- Redesigned a critical econometric workflow by porting logic from Stata to Python, utilizing vectorization to decrease execution time by over 99% (from days to under 30 minutes) while enabling parallel processing.
- Spearheaded the migration of optimization inputs from legacy Excel files to a SQLite backend by architecting a custom Streamlit interface as part of a broader effort to modernize resilient supply and logistics chains in complex environments; resolved critical infrastructure bottlenecks and received an internal award for rapid feature delivery.

Data Science Intern

- Designed and prototyped a multi-task convolutional neural network (CNN), developing a scalable data pipeline to train and integrate the model with large-scale linear optimization frameworks, reducing optimization runtime by over 80% (from 10+ minutes to under 2 minutes) through CNN-driven approximations of initial solutions.

Massachusetts Bay Transportation Authority (MBTA) | Boston, MA

June 2023 - August 2023

Transit Data Intern

- Developed a Python pipeline to batch-download and analyze historical MBTA prediction data from AWS S3, using Matplotlib visualizations of prediction accuracy to improve reliability of real-time information displayed across 20+ station and roadside signs during railway maintenance slow zones.

PROJECTS

Predictive Modeling of MBTA System Bottlenecks

- Developed a scalable cloud-based analytics platform integrating real-time transit data (MBTA V3 API), historical records, and weather data (OpenWeather API) to identify operational bottlenecks and predict service disruptions.
- Implemented data pipelines using Azure Data Factory and Databricks to preprocess and aggregate transit and weather data, facilitating exploratory analysis and predictive modeling.
- Built and evaluated machine learning models (Gradient Boosted Trees, Random Forests, Decision Trees, Linear Regression) to predict transit delays (headway/dwell time), achieving the highest accuracy (R^2 up to 0.81) with Gradient Boosted Trees, and identified significant impacts of weather conditions on service reliability.

Chain-of-Thought Reasoning Injection for LLMs

- Fine-tuned the DeepSeek-7B Large Language Model on the GSM8K benchmark using parameter-efficient Low-Rank Adaptation (LoRA) and QLoRA to engineer explicit Chain-of-Thought reasoning capabilities.
- Architected an automated grid-search pipeline to evaluate 108 hyperparameter configurations; identified optimal rank/alpha settings that stabilized training dynamics and minimized gradient norm spikes.
- Optimized training in a constrained cluster environment by implementing custom tokenization with full-sequence supervision and gradient accumulation, increasing exact-match accuracy from 5.0% to 38.0%.

Airbnb Pricing Prediction

- Developed and optimized predictive models (XGBoost, Random Forest, CatBoost) using a comprehensive NYC Airbnb dataset (48,000+ listings) to forecast rental prices based on property features, achieving an R^2 of approximately 0.64.
- Identified key determinants of pricing such as room type, neighborhood, and occupancy rate, providing actionable insights for Airbnb hosts to optimize rental strategies and improve market competitiveness.

EDUCATION

Boston University, Boston, MA

Collaborative Degree Program

B.S. in Data Science; B.A. in Economics

Relevant Coursework: Deep Learning, Machine Learning & Artificial Intelligence, Cloud Computing, Big Data Workflows, Economics of Risk and Uncertainty, Game Theory, Econometrics, Applied Statistics, Linear Algebra, Data Visualization.

SKILLS

Programming: Python, Rust, SQL, SPL, GAMS, LaTeX.

Libraries: pandas, transformers, scikit-learn, PyTorch, tensorflow, numpy, matplotlib, NLTK, geopandas.

Cloud & DBMS: AWS, Azure, Google Cloud Platform, Databricks, Splunk, SQLite, PostgreSQL, MongoDB.

Other Technical: Git, Docker, Kubernetes, Tableau, PowerBI, STATA, SAS JMP, Excel, MS Office.